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Problem Framing

Objective and Information Horizon

The primary objective of this study is to construct a binomial classifier to predict the **next-day directional movement** of the NIFTY 50 index. To ensure zero lookahead bias, the model adheres to a strict information cutoff. Here, I used Day T for the feature columns and Day T+1 Return as my Target Return

Initial Objective :

Using the 12 Selected Features of Day T, I would be predicting Day T+1 Return, which will be binarized as ("1" for positive return and "0" for non positive) using a Regressor Model

Updated Objective:

Using the 12 Features, the model will be classifying the Trend into "1" and "0" directly without involvement of return binarization using Xgboost Classifier Model.

Reason : After running multiple experiments, the Classifier Model outperformed the regressor model by a thin margin. The reason is likely due to the error increases while binarizing the predicted return instead directly classifying it.

Treatment of Flat and Near-Zero Returns

A critical design choice in directional forecasting is the handling of "flat" days where the return is near zero. In this model, **flat returns are bucketed into the Down (0) category**. **Alpha Threshold:** By labeling only positive returns as "1," the model is forced to identify signals strong enough to overcome the "friction" of the market. This creates a more conservative, and therefore more realistic, trading signal.

Baseline Performance

To evaluate the model's predictive edge, we establish three primary baselines calculated over the In-Sample window (January 2022 – June 2025). The baseline strategies are Always Up " and " Always Down" , (Reason) : Over a short duration, the Nifty trends usually have 50% chances of UP/DOWN

Insample_dataset	Result (win rate)
My_ Strategy	55.23%
Baseline_UP	53.21%
Baseline_DOWN	46.79%
outsample_dataset	Result
My_ Strategy	57.27%
Baseline_UP	52.73%
Baseline_DOWN	47.27%

Data Audit

Started with Correlation of each provided features with the Day T return (Same Day) then a second matrix to check every features with Day T+1 return. The correlation reduced for almost every features except **ma5_smooth_signal** , it just became opposite.

For cross checking I calculated all the given 32 features from scratch judging by their names,
3 Features with major issues:

I have calculated Error from my calculated features and starter_pack

	Feature	MAE	Max Error
0	ret_1d	2.166752e-05	1.174588e-02
1	ret_5d	1.406942e-04	1.797959e-02
2	ret_10d	3.158601e-04	1.773544e-02
3	ret_20d	6.205628e-04	2.164629e-02
4	ret_intraday	4.829972e-17	9.974660e-17
5	ret_overnight	2.155057e-05	1.171835e-02
6	high_low_range	6.392987e-05	2.457672e-03
7	log_volume	3.744338e-06	4.291753e-05
8	volume_ratio_20d	1.531899e-03	7.685191e-02
9	close_vs_ma5	3.493650e-05	4.517275e-03
10	close_vs_ma20	1.022658e-04	2.528249e-03
11	close_vs_ma50	1.740055e-04	2.971903e-03
12	momentum_5_20	6.645611e-04	2.291848e-02
13	vol_5d	1.086649e-01	5.733277e-01
14	vol_20d	1.187871e-01	2.813790e-01
15	vol_50d	1.209082e-01	2.235718e-01
16	rsi_14	7.304355e+00	2.905677e+01
17	bn_ret_1d	3.225164e-05	2.321380e-02
18	bn_ret_5d	1.809365e-04	2.339990e-02
19	nifty_bn_spread	1.519212e-05	1.146792e-02
20	nifty_bn_corr_20d	1.555682e-03	7.230550e-02
21	vix_level	0.000000e+00	0.000000e+00
22	vix_change	8.488663e-17	4.996004e-16
23	vix_5d_change	8.379017e-04	1.278445e-01
24	vix_ma_ratio	6.008388e-02	3.487654e-01
25	close_vs_252d_high	2.424677e-03	1.862030e-02
26	close_vs_252d_low	1.201389e-02	4.286261e-02
27	dow	0.000000e+00	0.000000e+00
28	ret_zscore	4.918077e-01	5.476316e+00
29	ma5_smooth_signal	3.748952e-03	2.832013e-02
30	volume_normalized	1.678405e-02	2.315873e-02

U can clearly see the significant difference (which MAE is in order greater then equal to 0.2)

Ret_zscore , volume_normalized , close_vs_252d_low , vix_ma_ratio , vol_(every) , which is showing a significant difference from the calculated , for which i have created few features from the OHLC data which includes supertrend , Bollinger Band and Gap

Feature engineering and EDA

Data I choose to keep: (9 features)

Close_vs_ma5 : Identifies short-term mean-reversion exhaustion when price overextends from its weekly average.15:30 IST (Day T)

Momentum_5_20 : Measures trend acceleration by comparing weekly performance against the monthly baseline.15:30 IST (Day T)

Nifty_bn_spread : Captures sector rotation and relative strength between the NIFTY 50 and the Banking sector.15:30 IST (Day T)

Nifty_bn_corr_20d : Monitors regime stability; decoupling often precedes structural index volatility.15:30 IST (Day T)

Vix_ma_ratio : Normalizes fear levels to identify volatility "shocks" relative to the recent 5-day average.15:30 IST (Day T)

Data from my own:

S_t_indicator: A volatility-adjusted trend signal (Supertrend) providing a cleaner signal-to-noise ratio than MAs.15:30 IST (Day T)

Bn_ratio_ret_5d : Tracks the momentum of the Bank-Nifty/Nifty ratio to capture institutional repositioning.15:30 IST (Day T)

Bb_width_velocity : Identifies "Volatility Squeezes" versus breakouts by measuring the rate of band expansion.15:30 IST (Day T)

Gap_pct : Quantifies overnight sentiment normalized by ATR, capturing reactions to global market cues.

(Why only 9 ? The additional features created random noise which disrupt the signal)

close_vs_ma5	1.00	0.02	-0.09	-0.06	-0.52	0.09	0.11	-0.02	0.48
momentum_5_20	0.02	1.00	-0.02	0.28	-0.03	0.36	0.01	0.05	-0.01
nifty_bn_spread	-0.09	-0.02	1.00	-0.01	0.09	-0.01	-0.47	-0.01	-0.09
nifty_bn_corr_20d	-0.06	0.28	-0.01	1.00	-0.01	0.41	0.02	0.09	-0.02
vix_ma_ratio	-0.52	-0.03	0.09	-0.01	1.00	-0.14	-0.10	0.08	-0.36
s_t_indicator	0.09	0.36	-0.01	0.41	-0.14	1.00	-0.03	0.06	-0.05
bn_ratio_ret_5d	0.11	0.01	-0.47	0.02	-0.10	-0.03	1.00	0.01	0.07
bb_width_velocity	-0.02	0.05	-0.01	0.09	0.08	0.06	0.01	1.00	0.03
gap_pct	0.48	-0.01	-0.09	-0.02	-0.36	-0.05	0.07	0.03	1.00
	close_vs_ma5	momentum_5_20	nifty_bn_spread	nifty_bn_corr_20d	vix_ma_ratio	s_t_indicator	bn_ratio_ret_5d	bb_width_velocity	gap_pct

Every selected feature is below a standard threshold of **0.5** , Upon reducing the Features which was collinear (Return_5d and Close_vs_ma5) The Win rate marginally improved

Win Rate my strat: 55.11%	full insample_data
always up strat: 53.21%	Win Rate my strat: 55.23%
always down strat: 46.79%	always up strat: 53.21%
	always down strat: 46.79%

MODEL

I was initially testing on two Xgboost model one was Classifier and other was Regressor

Where the Regressor Model (for predicting Return)

Uses Squared Error as loss function

Where the Classifier Model (for classifying Return)

Uses Logistics Regression

```
# for using binary target
model = xgb.XGBClassifier(
    n_estimators=50,
    max_depth=3,
    learning_rate=0.05,
    random_state=42,
    objective='binary:logistic'
)

model = xgb.XGBRegressor(
    n_estimators=50,
    max_depth=3,
    learning_rate=0.05,
    random_state=42,
    objective='reg:squarederror'
)
```

Parameter :

Estimators: 50 (Limited to prevent overfitting on small local windows).

Max Depth: 3 (Shallow trees are preferred to ensure the model learns generalizable rules rather than memorizing noise).

Learning Rate: 0.05 (A conservative rate to ensure stable convergence during daily refits).

Validation:

Walk Forward Validation with Window Size = 15 (Sliding Window)

Reason : for Predicting T+1 Day Label (Return) last three Trading week will be enough to determine a trend , increasing would make not significant difference in UP/DOWN trend , and Window Size less than 15 will create a problem in label classification , There can be cases with entire window size consist to only 1 label .The model would be biased in that case.

```
# sliding window
X_train = final_df.loc[i - window_size : i - 1, feature_cols]
y_train = final_df.loc[i - window_size : i - 1, target_col]

# Predict for the current day (i)
X_test = final_df.loc[[i], feature_cols]
y_test = final_df.loc[i, target_col]
```

Result

Insample dataset:

My Strat	Always UP	Always DOWN
55.11%	53.21%	46.79%

OutSample

56.36%	52.73%	47.27%
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For an better statistical significance i divided Insample_dataset into **10 fold sequentially** , each trained separately to understand towards model inclination towards UP/DOWN

Strat	56.34	59.15	46.89	60.56	60.56	60.5	49.3	64.21	48.43	41.86
UP	40.85	57.75	54.93	56.34	61.97	61.97	57.75	57.14	40	57.14
Down	59.15	42.25	45.07	43.66	38.03	38.03	42.25	42.25	60	42.86

The Model slightly perform better in bullish regime. Even infact in bearish regime chunk 9 , strat 51.43% time correctly identified trend despite bearish returns. Most of the chunks it managed to stay above 50%

Outsample dataset Metrics

HitRate (**56.32%**) & AUC (**0.5601**) correctly predicting the NIFTY's direction ~56% of the time. In the Indian market, a 6% edge over a coin flip is significant.

Balanced Accuracy (56.01%): This is the "no-bias" proof. Since it is identical to your AUC, it proves model is equally skilled at calling "Down" days as it is "Up" days.

The "Honesty" Check (CI): Lower Confidence Interval (0.46) is below 0.50. You **must** mention in your report that while the *average* performance is strong, the sample size is small enough that we cannot fully rule out luck yet.

	Score	95% CI (Low)	(high)
AUC	0.5711	0.47	0.66
Bal_Acc	0.5609	0.47	0.66
Sharpe	2.8532	-0.19	5.8
Hitrate	0.5742	0.4727	0.6545
turnover	0.4953	0.403	0.587

Negative CI suggest Bootstrapping on a small sample might creates high variance.

How do I know this edge is real?

The model achieved a mean hit rate of **56.42%** in the out-of-sample window. However, the 95% Confidence Interval (CI) of **[0.4727 to 0.6545]** indicates that the lower bound remains below the 50% "random walk" threshold.

The Verdict: While the mean performance suggests a valid signal, we cannot claim 95% statistical confidence that the edge is permanent. This is a common "small sample size" effect in 6-month windows. To confirm the edge, I would require a longer out-of-sample period (minimum 12-18 months) to narrow these intervals.

The in-sample "chunking" (dividing the data into 10 sequential folds) proved that the model's edge is **regime-dependent**.

- In **Chunk 1 and 9**, the model successfully outperformed the "Always Up" baseline during bearish periods.
- In **Chunk 3 and 10**, the model underperformed.
- **The Insight:** This variance proves the model isn't just biased toward the NIFTY's long-term drift. It shows the model is actively attempting to find structural signals, even if it occasionally fails during regime transitions. This justifies the **15-day walk-forward refit cadence** to keep the model "fresh."

Survival against Random-Label Baselines :

Because the **Balanced Accuracy (56.09%)** is identical to the **Hit Rate**, it confirms the results survive a "random-label" comparison. The model is not gaining accuracy by over-predicting the majority class; it is gaining accuracy by correctly identifying both up and down regimes.

Reality :

Transaction Costs (STT/Slippage): With a **Turnover of 0.49**, the strategy trades every two days. In India, round-trip costs (including STT) could eat **3-5% per year**. My Sharpe of 2.85 is a **gross** figure and would likely compress to **1.5 - 1.8** in a live environment.

The "Shorting" Gap: The **Sharpe CI lower bound of -0.19** reveals that in a Long/Short framework, the model is vulnerable to overnight gaps. A 2% gap-up against a Short position would be more damaging than the backtest currently simulates.

In Another Week , I will implement Position Sizing Scale down bets when **vix_percentile** is in the 90th percentile to protect against tail risk.and research **ATR-based Stop Losses**, To fix the drawdown risk identified in the Long/Short simulation.